

HRRP /DUDBC

Concrete Block and CSEB Masonry building



COMPRESSED STABILIZED EARTHEN BLOCKS (CSEB)

AN ALTERNATIVE CONSTRUCTION MATERIAL

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Resilient Community through SAFER Infrastructure



BACKGROUND

BUILDINGS construction & material contribute as much as **one third** of total **global greenhouse gas emissions**, primarily through the use of fossil fuels during their operational phase.

(Intergovernmental Panel on Climate Change (IPCC))

High Embodied energy
+
Use of much Fossil Fuel
+
High Carbon Emission



Deforestation
and
Loss in National Economy



Because of :

Building Material

Kiln burnt bricks
Cement

Indoor Energy Consumption

Heating / Cooling
Lighting
Ventilation

Construction

Transportation

Global Concern



Poverty

Global Warming

Pollution

Disaster

Illiteracy

Resource Crisis

Climate Change

Health

Energy Crisis

Environmentally
responsible
material
sourcing

A disaster hot spot

BACKGROUND CORD

4

6th to climatic hazards

(UNDP-2004 “Reducing Disaster Risk”), now it’s 4th

2

Growing climate variability

Mean annual temperature increase by: 0.5-2.0°C (2030s) & 1.7-4.1°C (2060s)

3

The conventional brick-concrete buildings practices growing

Do not perform well in severe temperatures.

The retrofitted measures require high cost materials and skilled labor which are largely unaffordable by the poor

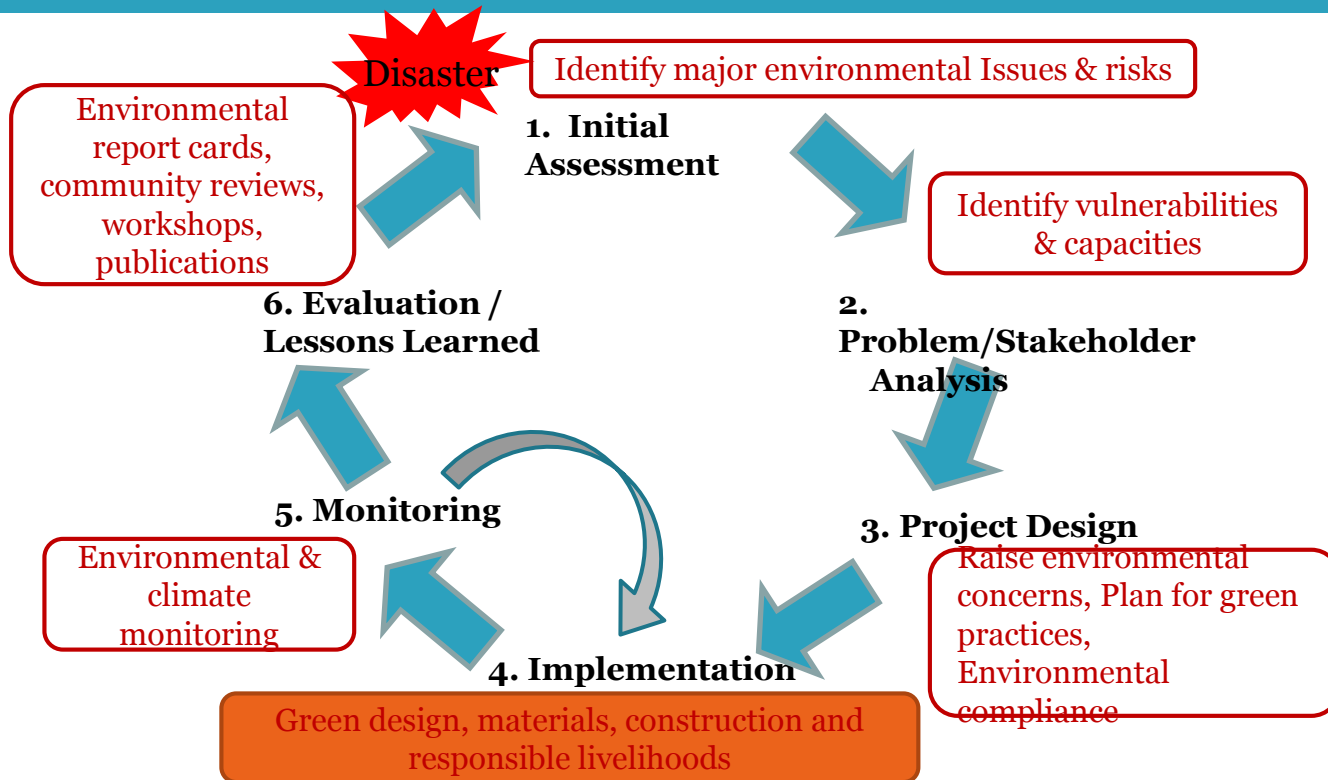


From **GREEN** towards

GREY

Project Cycle and Environmental Entry Points

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Alternative solution – Reconstruction of Gorkha Earthquake 2015

HENCE

It requires appropriate, affordable,
energy efficient, environmental
friendly, climate responsive, Disaster
resilient building solutions



Alternative Solution

Green Technology

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indigenous construction technology

use of local material

less embodied energy and
environment friendly

Assures comfort and
human health

Proper utilization of natural
resources

*Great Wall of China (3000 years
ago),
The Horyuji Temple in Japan
(1300 years)*



*Earth as Construction Material:
ADOBE or SUN DRIED BRICKS*

Began more than 9000 or over 10,000
years ago

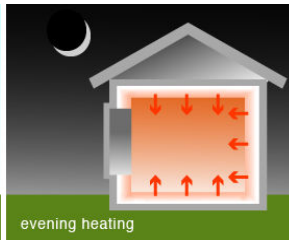
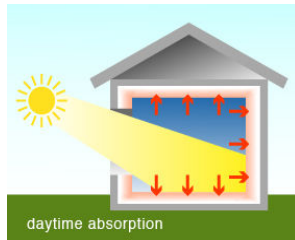
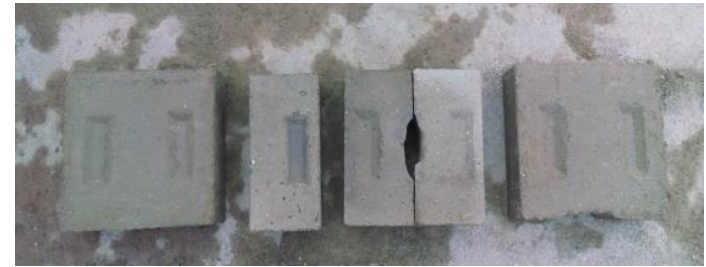
Earth construction - an environmental advantage
that makes it extremely competitive comparing to
conventional materials and construction techniques

CSEB – a green technology

CSEB – Compressed Stabilized Earthen locks

- alternative material to fired brick for building construction

- Environment Friendly
- Climate Responsive
- Community Employment
- Cost Effective
- Earthquake Resistant
- Livelihood/ local economy



CSEB - GREEN TECHNOLOGY

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Advantages?

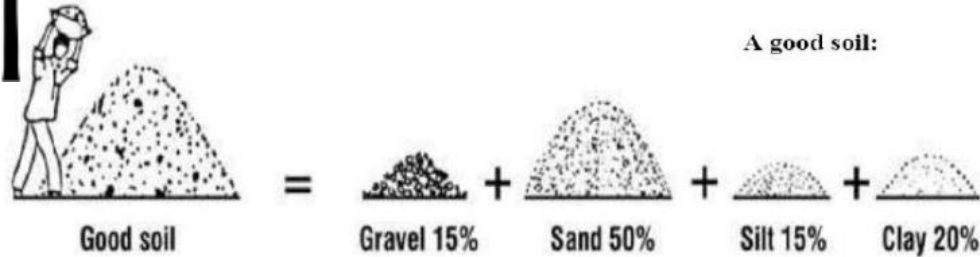
- Environment Friendly – Eco friendliness
- Climate Responsive – Energy Efficiency
- Livelihood – Labour based technology
- Economic Benefit – Direct and Indirect
- User friendly technology - Advanced technology on traditional practice
- Social acceptance – Community Participation
- A bio-degradable/local and Sustainable material
- Disaster Resilient/ FIRE RESISTANT

Compressed Stabilized Earthen Block - CSEB
As Alternative material

**Alternative
technology**

Ecological Performance of CSEB

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CSEB has these advantages compared to fired bricks:

Pollution emission(Kg of CO ₂ /m ²)	Energy consumption (MJ)
7.9 times less than country fired bricks	15.1 times less than country fired bricks

Ecological comparison of building materials				
Product and thickness	No of Units (per m ²)	Energy consumption (NJ per m ²)	CO ₂ emission (Kg per m ²)	Dry compressive strength (Kg/cm ²)
CSEB-24 cm	40	110	16	40-60
Wire Cut Bricks-22 cm	87	539	39	75-100
Country Fired bricks-22cm	112	1657	126	30-100
Concrete blocks-20 cm	20	235	26	75-100

Note: Wire Cut bricks are also called Kiln fired bricks. (Source: Development Alternatives 1998)

**Environment
Friendly
Advantages**

Climate Responsive

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Control Relative Humidity than ventilation

**Indoor and out door temperature difference –
Approx. 5 deg centigrade**

Advantages

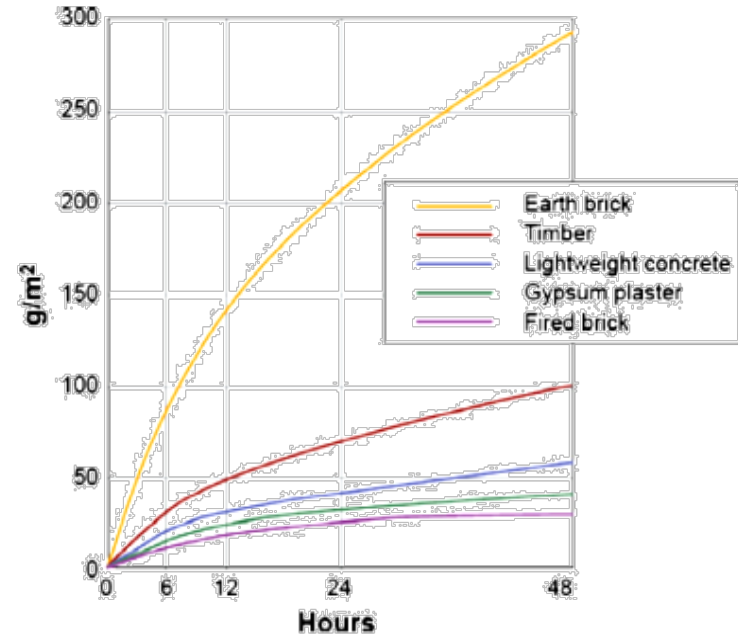
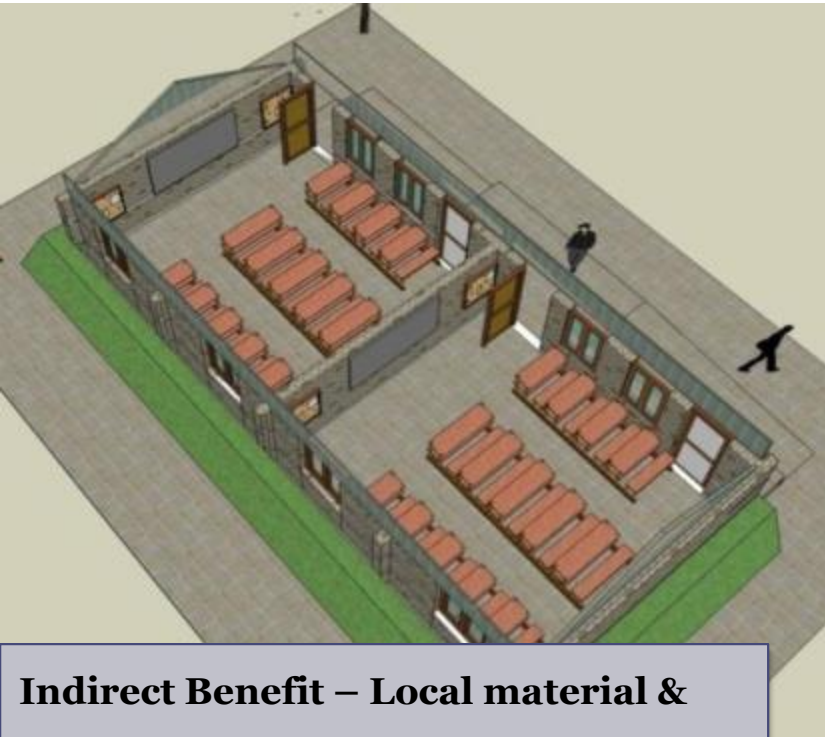


Fig: Weight of moisture absorbed by different materials when relative humidity increases from 50% to 80%

Economic Benefit

Advantages

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Construction cost and time

Building with Plinth Area = 880 sq. ft

Block Production per day = 600

Estimated no of blocks required
= 6000 nos.

Total days for block production with 21 days for curing =
31 days

**Total Building construction time
= 3-4 months**

**Estimated cost for 2 rooms one storey
building with 880 sq ft plinth Area
= Approx NRs 1 million**

Rs 1200/sq. ft

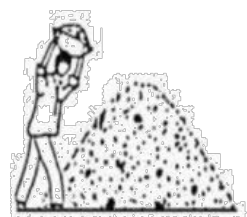
**Direct economic benefit 10 % less than
Brick masonry building of same plinth
area**

**Indirect Benefit – Local material &
labor based**

User Friendly Technology Advantages

Advanced technology on traditional practice

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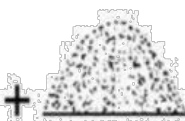


Good soil

=



Gravel 15%



Sand 50%



Silt 15%



Clay 20%

A good soil:

Stabilizer

Cement - for sandy soils 5% - 7 %

Lime - for very clayey soil

Advantages

Community Participation

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**Social Acceptance &
Bio Degradable**



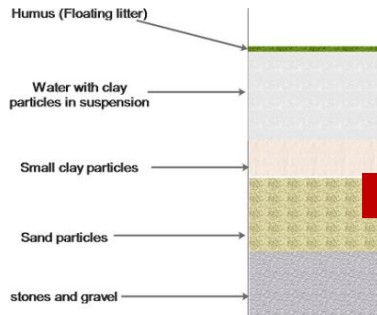
Local and Sustainable material



1. SOIL COLLECTION



2. SOIL TEST



3. BREAKING & SEIVING



7. CURING



8. CSEB BLOCKS



4. PROPOTIONING



6. MOULDING



5. MIXING

CSEB PRODUCTION

Capacity Building
through training and
technology transfer

Support local economy

Labor Based

GUIDELINES and PLANS on CSEB developed by CoRD/AAN and endorsed by DUDBC



- Standard Norms and Specification for CSEB Block, July 2012
- Hand Book on Community Engagement, July 2012
- Manual for Earthquake Safe Building, July 2012
- Manual For Engineer, July 2012

Analysis & Design of CSEB Masonry Building

Base shear Calculation

Zonal factor, $Z=0.36$

Importance factor, $I=1$

Response reduction factor, $R=1.5$

Considered height, $H=6.072$

Length along x-direction, $L_x=8.905$ m

Length along y-direction, $L_y= 5.91$

Time period, $T_x=0.183$ sec

$T_y=0.225$ sec

$$S_a/g_x=S_a/g_y=2.5$$

$$A_h=z/2 \cdot I/R \cdot S_a/g=0.09$$

Sesmic wt. of the building, $W=1308.5$ KN

$$\text{Base Shear} = A_h \times W = 392.55 \text{ KN}$$

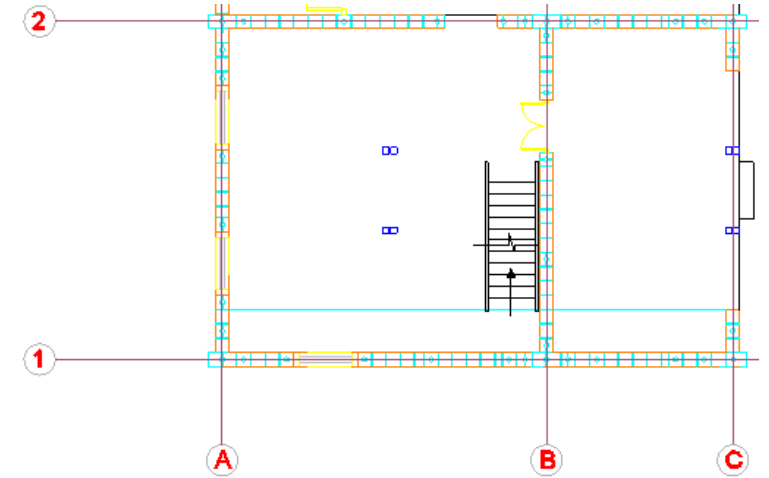
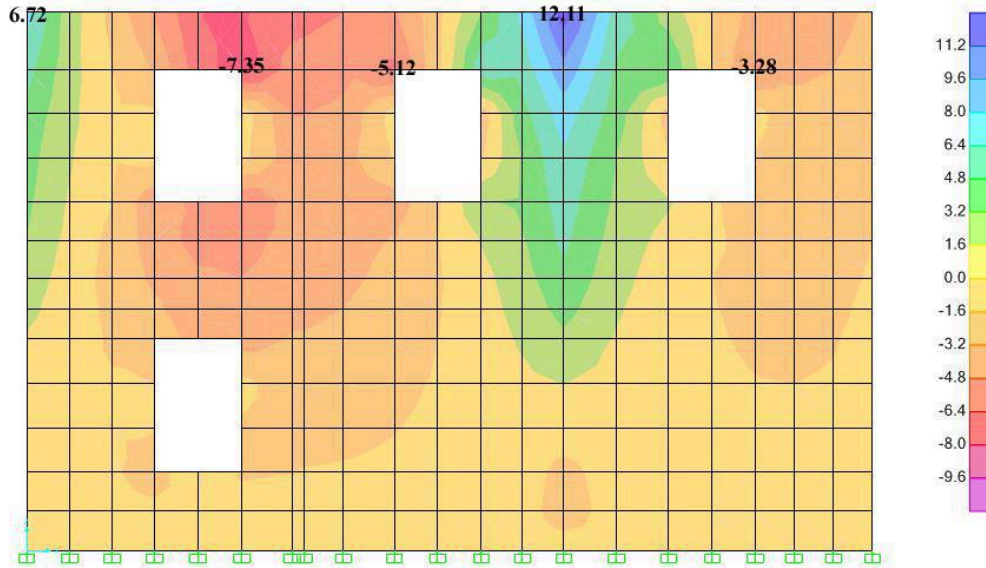
NBC -

SAP

STRUCTURAL CALCULATIONS of CSEB

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Out of Plane Bending

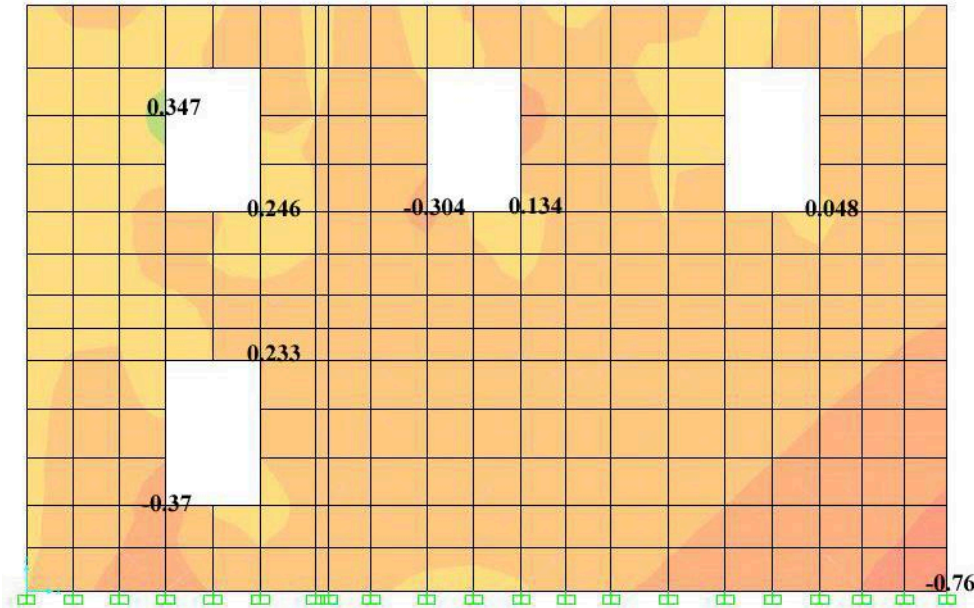


Grid Line of the building
For Grid 1-1

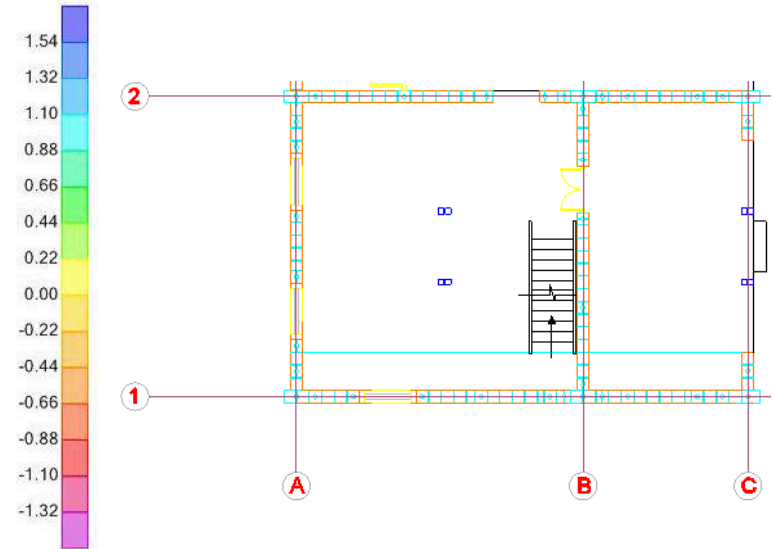
STRUCTURAL CALCULATIONS of CSEB

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Compressive and Tensile Force in Plane Bending



For Grid 1-1



STRUCTURAL CALCULATIONS of CSEB

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Check for Shear Stress

wall ID 1 for load combination EQ_x

Total shear force in the wall = 168.47 KN
 Length of the wall = 8.905 m
 Thickness of the wall = 240 mm
 Shear stress in the wall = 0.078827 N/mm²
 Assuming total shear force to be resisted by the wire mesh in horizontal and vertical bands.

Considering with reinforcement of Fe415

4.75mm@150mm c/c, f_y=415 wire mesh vertical band in both inner and outer faces of the wall.

shear strength per meter of the vertical bands = 43.82 KN/m

shear strength of 225 mm splint band in inner face 9.8595 KN

Number of vertical bands along the inner face of the wall = 2

Total shear strength of outer band and inner vertical bands $8.905 \times 43.82 + 9.8595 \times 2$
 = 409.9361

Minimum expected shear strength of masonry = 0.1 N/mm²

Total shear strength of masonry unit = 213.72 KN

Now, ratio of shear strength to shear force = $(409.9361 + 213.72) / 168.47$
 = 3.70 ok

CONSTRUCTION SEQUENCES OF CSEB BUILDING



Step 01: Foundation trench excavated to hard ground



Step 02: Stone soling with sand in foundation

CONSTRUCTION SEQUENCES OF CSEB BUILDING



**Step 03: Vertical Reinforcement
with PCC in foundation**



**Step 04: Completed foundation ring beam with RCC
and Vertical Reinforcement**

CONSTRUCTION SEQUENCES OF CSEB BUILDING



Step 05: CSEB lay above foundation ring beam



Step 06: U-block with reinforcement in plinth band

CONSTRUCTION SEQUENCES OF CSEB BUILDING



Step 07: Completed concreting of plinth beam



Step 08: Installation of Door

CONSTRUCTION SEQUENCES OF CSEB BUILDING



Step 09: U-block with reinforcement for sill band



Step 10: Window Installation with Vertical Reinforcement on its either side

CONSTRUCTION SEQUENCES OF CSEB BUILDING



Step 11: U-block with reinforcement in lintel band



Step 12: Completed Structure up to lintel band after concreting lintel band

CONSTRUCTION SEQUENCES OF CSEB BUILDING



ROOF BAND

LINTEL BAND

CORNER BAND

SILL BAND

PLINTH BAND

Step 16: Important structural elements marked in completed CSEB model

PROJECTS on CSEB Masonry

**TECHNOLOGICAL DEMONSTRATION CENTER, GORKHA – UNDP,
CoRD**

GREEN SCHOOL PROJECT– AAIN, CoRD
➤ **16 School Building and 3 ECD**

PROJECTS BY CoRD on CSEB

TECHNOLOGICAL DEMONSTRATION CENTER, GORKHA – UNDP, CoRD



In order to promote the construction technology on CSEB, UNDP with technical support from CoRD, constructed CSEB building in Gorkha- ***“Build Back Better”***

PROJECTS BY CoRD on CSEB

TECHNOLOGICAL DEMONSTRATION CENTER, GORKHA – UNDP, CoRD COST EVALUATION

Building Construction Cost Evaluation

Built up Area: 300sq.ft. (100 sq.ft. balcony)

Description	Till Plinth Level	Plinth till Roof	Roof	Total
Construction Material	83000	266000	112000	461000
Labor Cost	44000	124000	38500	206500
Total Cost	127000	390000	150500	667500
Government Support	50000	150000	100000	300000
House Owner Expense	77000	240000	50500	367500

Construction Cost Per Sq. ft. : Rs. 2,200/-

PROJECTS BY CoRD on CSEB

TECHNOLOGICAL DEMONSTRATION CENTER, GORKHA – UNDP, CoRD QUANTITY OF MATERIALS REQUIRED

Construction Materials	Till Plinth Level	Plinth till Roof	Roof
Stone (Trip)	1.8	-	-
Soil(Trip)	2.3	-	-
Wood(Trip)	1.5	0.2	-
Aggregate(Trip)	0.8	1.6	-
Sand(Trip)	0.8	2	-
Cement (Bags)	35	80	-
Rebars (Kg)	175	190	-
CSEB (Numbers)	-	2350	-
Pipe (Hand)	-	-	181
CGI Sheets (Bundle)	-	-	3.5

NOTE:

1 trip = 3 sq.m

1 Hand = 0.45 m = 1'6"

Sustainable green design and construction practices- Green School Project AAN/CoRD- Example



Model Building, (IOE)



School Building Morang



School Building Dolakha

16 School, 1 Model
and 3 ECD Building

Thermal Insulation – CGI Sheet Roofing



**Climate
Responsive**



**Thermal adaptation Research
Work**



**CS-board as alternate
material for false ceiling**



Straw-bamboo insulation

Green Technology School

Performance on Gorkha Earthquake 2015 Kutl Danda Dolakha



Green Technology School

Performance on Gorkha Earthquake 2015 Mirge Dolakha



Green Technology School

Performance on Gorkha Earthquake 2015 Sailungeswor Dolakha



Green Technology School

Performance on Gorkha Earthquake 2015 Singati Dolakha

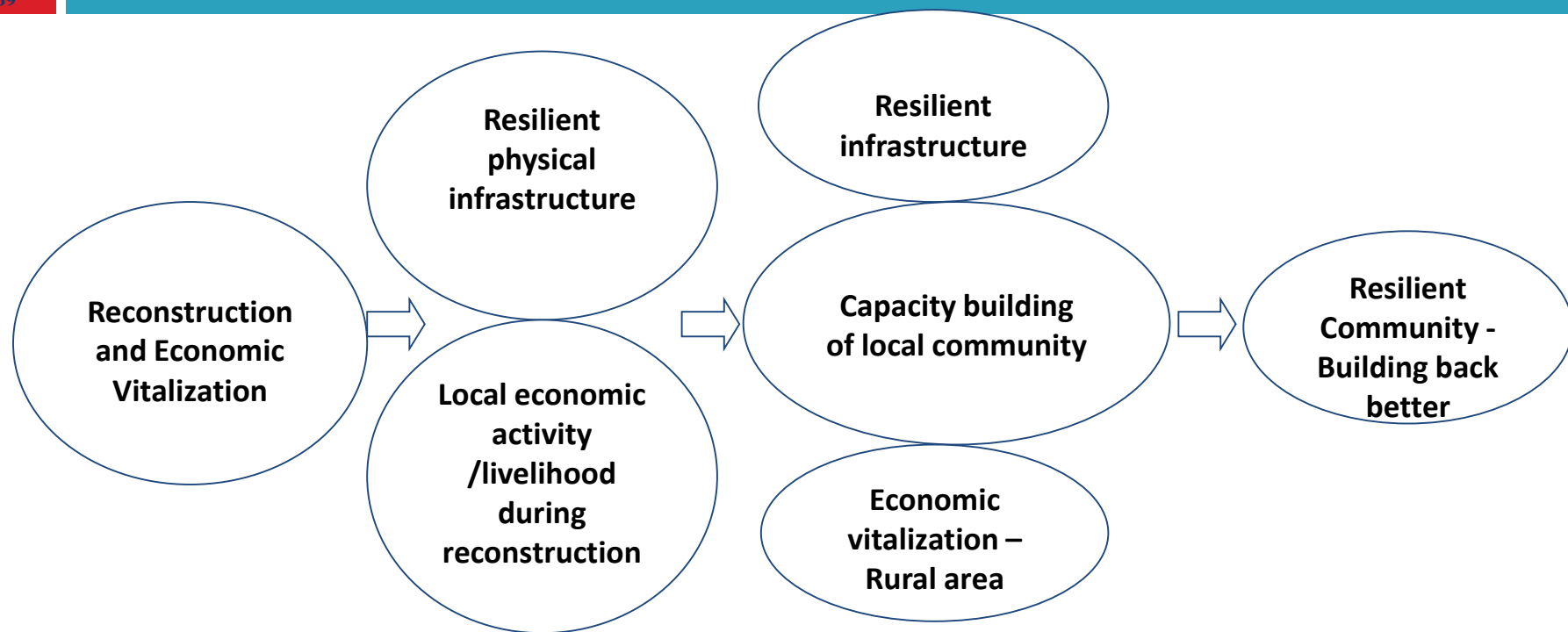


Green Technology School

Performance on Gorkha Earthquake 2015 Kuti Danda Dolakha



Build Back Better Concept



Resilient Community through Resilient Infrastructure

THANK YOU